

EDUCATIONAL PROJECT PRESENTATION

Design & Implementation of a Hotel Services DBMS

One source of truth—from reservation to revenue.

PRESENTER — SEPTEMBER 4, 2026

Disconnected systems create costly friction

DOUBLE BOOKINGS

INCORRECT BILLING

SERVICE DELAYS

INTEGRATED DBMS

Reservations, guest services, housekeeping and billing coordinated in real time.

AIM • Improve data accuracy, operational coordination, and service quality.

What the system owns—and what it enables

SCOPE

Room types · Rooms · Rate plans · Guests
· Reservations · Stays · Services ·
Housekeeping · Billing

OBJECTIVES

01

Relational design
Integrity + normalization

02

Operational SQL
Actionable reports

03

Working prototype
Core hotel workflows

Four perspectives. One coordinated workflow.



FRONT DESK

Search availability + book



HOUSEKEEPING

Assign and close tasks



BILLING & ACCOUNTS

Invoice + process payment

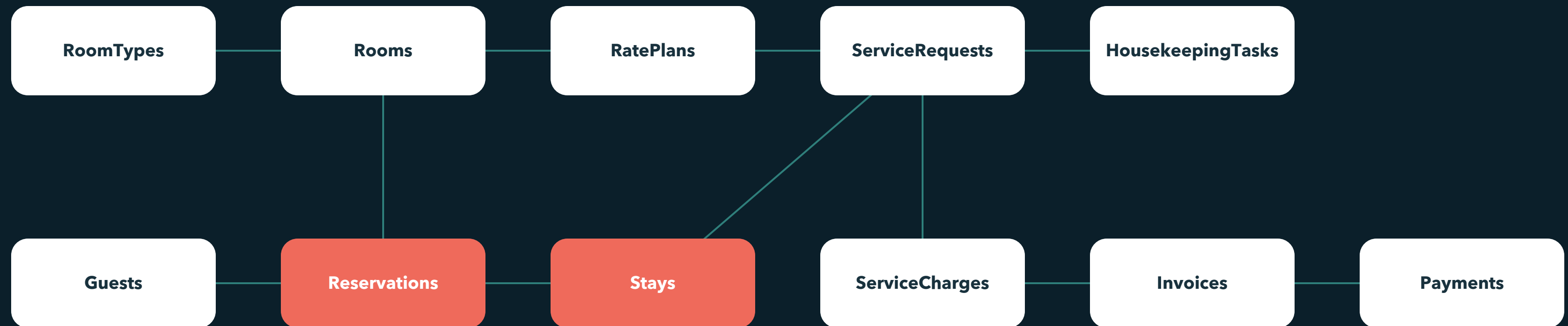


GUEST SERVICES

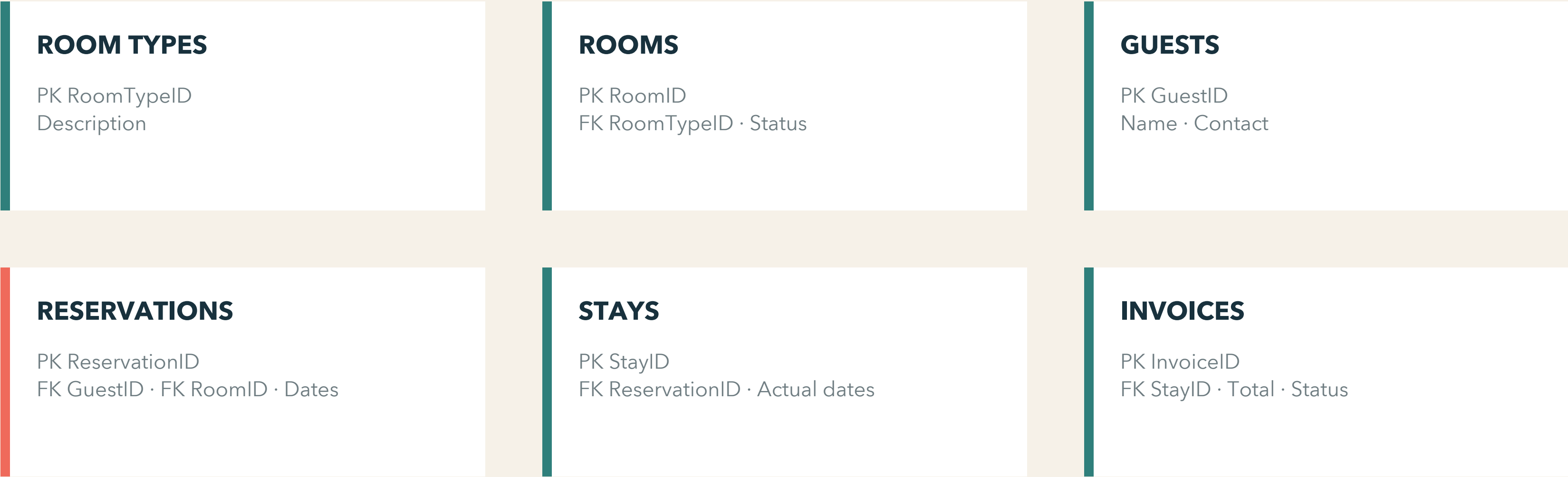
Post requests + charges

BOOK → ARRIVE → SERVE → SETTLE

Eleven entities connect the guest journey



Initial relational schema



RATIONALE · Stable identifiers, explicit relationships, and modular transaction tables keep change controlled.

From raw records to trustworthy 3NF

1NF

Atomic fields

Repeating service entries removed

2NF

Whole-key dependence

Partial dependencies eliminated

3NF

No transitive dependence

Reservations $\neq$ Stays; charges $\rightarrow$ InvoiceID

Business rules enforced at the data layer

NO OVERLAPS

A room cannot host concurrent stays

TRIGGER

VALID DATES

Check-out must follow check-in

CHECK

CAPACITY

Guests cannot exceed room capacity

CHECK + FK

SAFE MONEY

Charges non-negative; payments capped

CHECK

Operational questions, answered in SQL

78%

TODAY'S OCCUPANCY

VACANT ROOMS
12 ready · 4 cleaning

SERVICE REVENUE
\$8,420 MTD

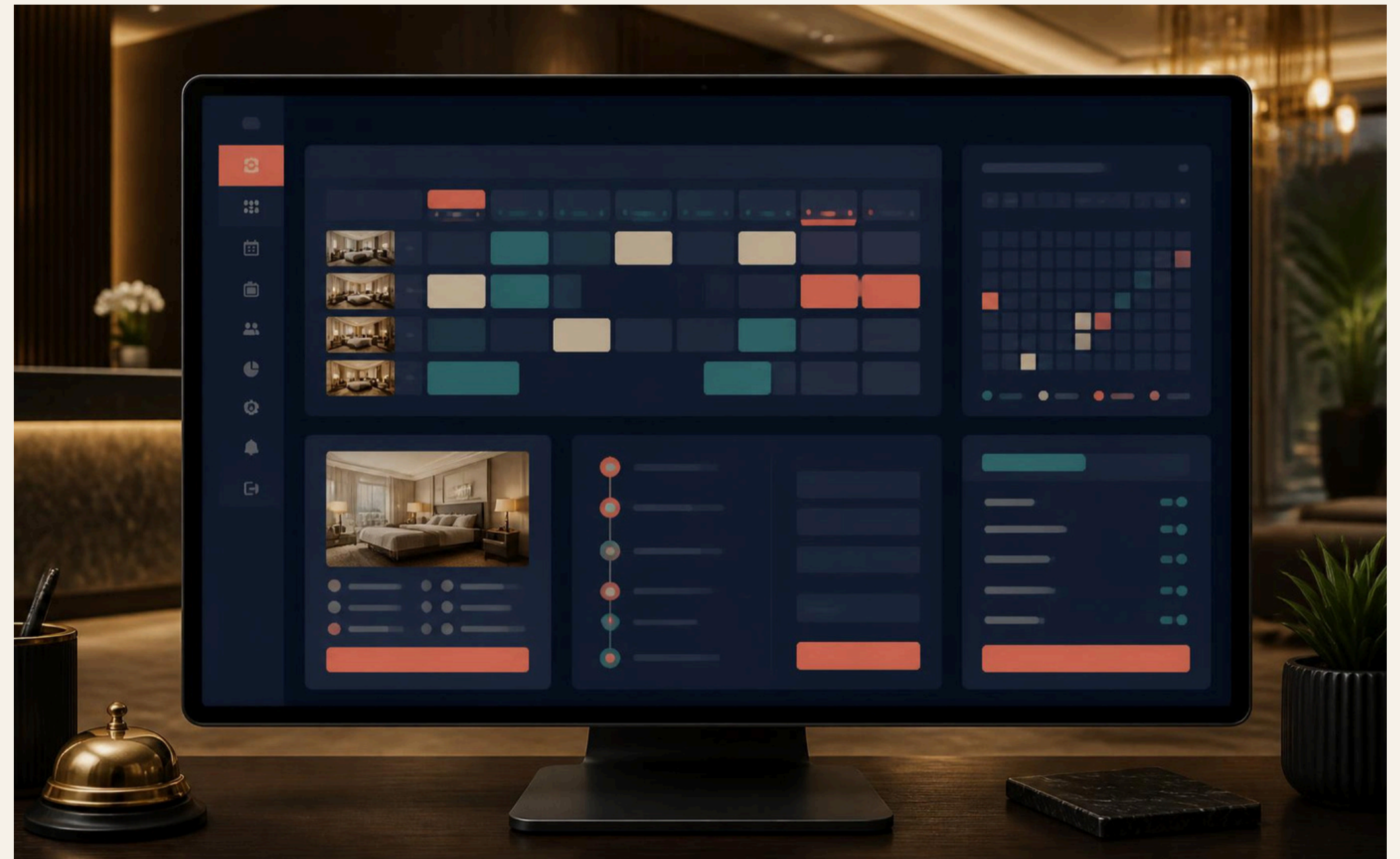
```
SELECT room_type,  
COUNT(*) AS occupied  
FROM stays  
WHERE :today BETWEEN check_in AND  
check_out  
GROUP BY room_type;
```

Also: cancellations · housekeeping status · guest history · total revenue

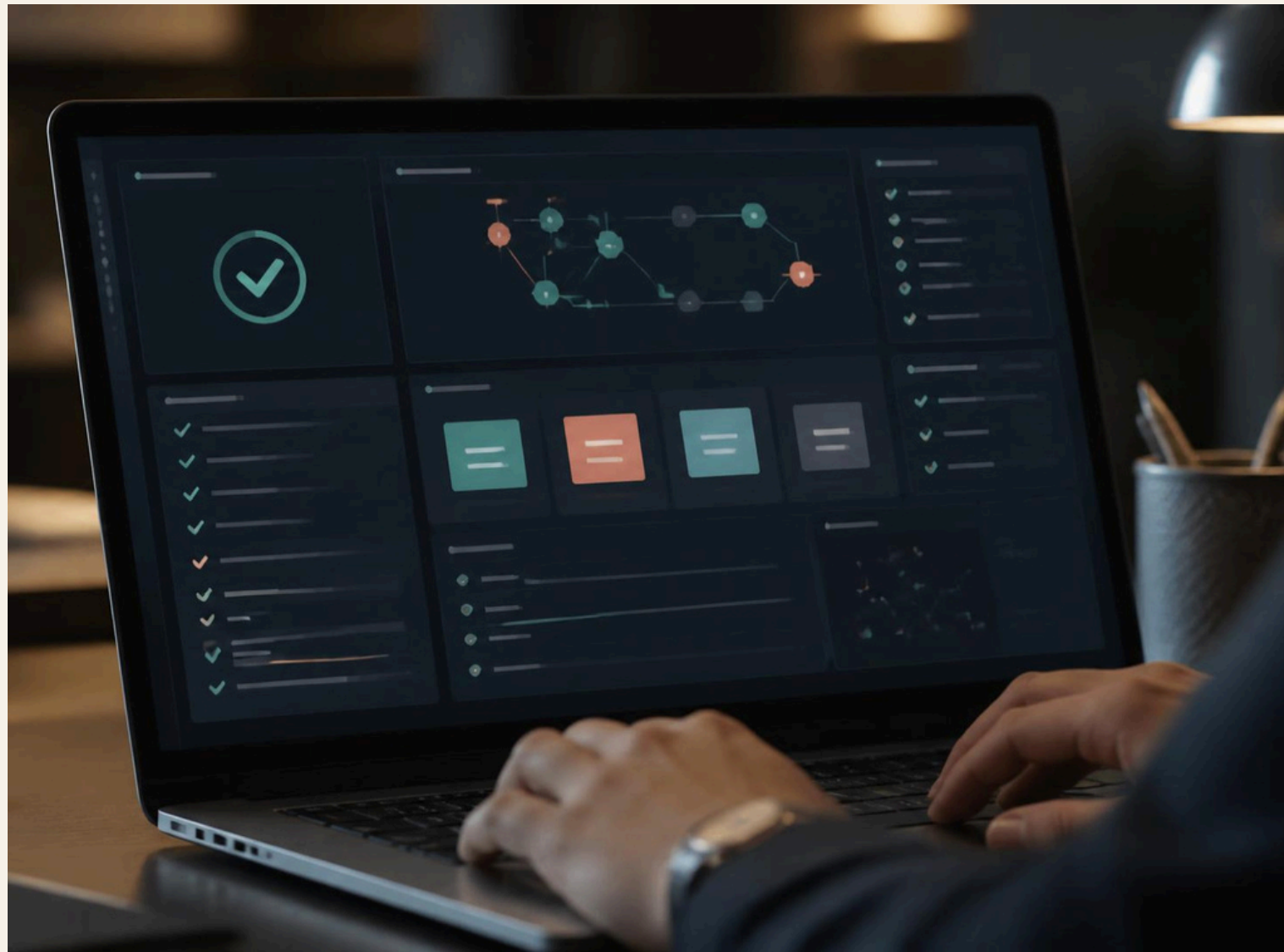
Core hotel workflows in one application

PYTHON / JAVA + SQL

Search & book
Check in / out
Post services
Assign housekeeping
Invoice & pay
Management reports



Test the workflow—and the failure modes



01 **CRUD + field validation**

02 **Transaction rollback + errors**

03 **Booking → arrival → billing**

Foundation approved

01

Problem · Scope · Objectives
Users · Functional requirements
Conceptual ER diagram
Initial relational schema

Database made dependable

02

3NF + dependencies
DDL constraints + business rules
Sample data + SQL reports
Implementation progress

From database to demonstrable product

03

Completed application
CRUD · search · validation · reports
Source · scripts · test data · screens
Project report + viva-voce

CONCLUSION

One integrated system. Every guest touchpoint.

A hotel management DBMS improves accuracy, coordination, and operational confidence—from first search to final payment.

Questions?